



BREAST CANCER CLASSIFICATION FROM HISTOPATHOLOGICAL IMAGES USING CONVOLUTIONAL NEURAL NETWORK IN DEEP LEARNING

MINI PROJECT WORK

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JANUARY 2022

This is to certify that this project work entitled

**BREAST CANCER CLASSIFICATION FROM
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LEARNING**

is the bona-fide record of project work done by

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SYNOPSIS

Deep learning is an artificial intelligence (AI) function that imitates the workings of the human brain in processing data and creating patterns for use in decision making. Several deep learning algorithms have been recently introduced to scientific communities and are applied in various application domains. In this project, Convolutional Neural Network [CNN] is used for image classification. There are mainly three things in CNN: i) local receptive field, ii) shared weight and biases, iii) activation and pooling. In CNN, first the neural networks are trained using a heavy set of data so that the CNN can extract the feature of given input. When the input is given, first image preprocessing is done then the feature extraction occurs on the basis of set of data stored and then the classification of data is done and output is shown as the result. The main aim of our project is to detect and classify the histopathological images as cancerous or non-cancerous using convolutional neural networks.

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LIST OF ABBREVIATIONS

ANN	ARTIFICIAL NEURAL NETWORK
CAD	COMPUTER AIDED DESIGN
CC	CRANI CAUDAL VIEW
CNN	CONVOLUTIONAL NEURAL NETWORK
EX-DBC	EXPERT SYSTEM DUCTAL BREAST CANCER
GMMs	GAUSSIAN MIXTURE MODELS
GPU	GRAPHICS PROCESSING UNIT
KNN	K NEAREST NEIGHBOUR
MLO	MEDIO LATERAL OBLIQUE VIEW
MLP	MULTI LAYER PERCEPTRON
OPEN CV	OPEN SOURCE COMPUTER VISION
PCA	PRINCIPLE COMPONENT ANALYSIS
RAM	RANDOM ACCESS MEMORY
RF	RANDOM FOREST
SVM	SUPPORT VECTOR MACHINE
WAUCE	WEIGHTED AREA UNDER THE RECEIVER OPERATING CHARACTERISTICS CURVE ENSEMBLE

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Breast cancer can originate from any cell, tissue, or gland of the breast. Sometimes it starts from the ducts that produce milk and sometimes it originates from the lobules which are glandular tissues. If not diagnosed at an early stage there is a possibility of spreading the cancer cells toward different parts of the body and cause damage there too. Breast tumor has two most common types: benign and malignant where the benign lesion is not cancerous, it is some kind of abnormalities in the cell and they are unable to become a cause of breast cancer and malignant is cancerous lesions. Malignant cells spread at a very fast rate by start divisions swiftly because both cells (benign and malignant) have irregular appearance and structure, it is a very difficult task to manually analyze the microscopic image. To overcome this issue and making this process completely automatic deep learning has been introduced. CNN has been used in many types of researches in medical image processing but used very rarely in breast cancer classification. In this study, application of deep learning has been exploit using the CNN model for the classification of breast cancer using histopathology images.

1.2 EXISTING SYSTEM

Several methods can be used to diagnose Breast Cancer, including breast biopsy, ultrasound, mammography, thermography, and fine needle aspiration cytology which requires the availability of pathologist for diagnosis which might reduce the accuracy rate. In order to improve breast cancer outcomes and survival, early detection is critical. Since detection of cancer is key to effective treatment of breast cancer we use various machine learning algorithms to predict if a tumor is benign or malignant, based on the features provided by the data. The limitation of most of

the machine learning study is the lack of experimental data because of which the cost of the experiment increases. More accuracy could be achieved by the better labeling of the input images and by using high-quality data.

1.3 PROBLEM DEFINITION

According to the world health organization (WHO) Breast cancer is the most frequent cancer among women, impacting 2.1 million women each year, and also causes the greatest number of cancer-related deaths among women. In 2018, it is estimated that 627,000 women died from breast cancer-that is approximately 15% of all cancer deaths among women. While breast cancer rates are higher among women in more developed regions, rates are increasing in nearly every region globally. Accurate detection and classification of breast cancer is a critical task in medical imaging due to the complexity of breast tissues. Due to automatic feature extraction ability, deep learning methods have been successfully applied in different areas, especially in the field of medical imaging. A deep learning method, Convolutional Neural Network model for the Breast Cancer Classification with histopathological images is proposed.

1.4 PROPOSED SYSTEM

The proposed model is working in an unsupervised fashion for the extraction of features from the input histopathology image patches in the form of feature vectors. The extracted features matrix is then transferred to the backpropagation neural network which is a supervised learning phase and it comprises of the conjugate gradient. A model is formed by the feature matrix of images in the training phase and the final stage is the classification stage which discriminate between cancerous and non-cancerous regions.

1.5 ORGANIZATION OF THE PROJECT

- Literature reviews of already existing proposals are discussed in chapter2.
- Chapter 3 has system specification which tells about the software requirements.

- Chapter 4 discusses the overall project and design which tells the brief description of each of the modules in this project.
- Chapter5 has the implementation and experimental result of the project.
- Chapter 6 deals with the conclusion and future work.
- Finally chapter 7 deals with the references.

CHAPTER 2

LITERATURE REVIEW

2.1 A SUPPORT VECTOR MACHINE BASED ENSEMBLE ALGORITHM FOR BREAST CANCER DIAGNOSIS

2.1.1 DESCRIPTION

This research studies a support vector machine (SVM)-based ensemble learning algorithm for breast cancer diagnosis. Illness diagnosis plays a critical role in designating treatment strategies, which are highly related to patient safety. Nowadays, numerous classification models in data mining domains are adapted to breast cancer diagnosis based on patients' historical medical records. However, the performance of each algorithm depends on various model configurations, such as input feature types and model parameters. To tackle the limitation of individual model performance, this research focused on breast cancer diagnosis that uses an SVM-based ensemble learning algorithm to reduce the diagnosis variance and increase diagnosis accuracy. Twelve different SVMs, based on the proposed Weighted Area Under the Receiver Operating Characteristic Curve Ensemble (WAUCE) approach, are hybridized. To evaluate the performance of the proposed model, Wisconsin Breast Cancer, Wisconsin Diagnostic Breast Cancer, and the U.S. National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program breast cancer datasets have been studied. The proposed WAUCE model reduces the variance by 97.89% and increases accuracy by 33.34%, compared to the best single SVM model on the SEER dataset.

2.1.2 MERIT

The experimental results show that the WAUCE model achieves a higher accuracy with a significantly lower variance for breast cancer diagnosis compared to five other ensemble mechanisms and two common ensemble models, i.e., adaptive boosting and bagging classification tree.

2.1.3 DEMERIT

One limitation of this study is that it requires a large amount of time and cost. This method is quite expensive.

2.2 IMPROVED MAMMOGRAPHIC CAD PERFORMANCE USING MULTI-VIEW INFORMATION: A BAYESIAN NETWORK FRAMEWORK

2.2.1 DESCRIPTION

Mammographic reading by radiologists requires the comparison of at least two breast projections (views) for the detection and the diagnosis of breast abnormalities. Despite their reported potential to support radiologists, most mammographic computer-aided detection (CAD) systems have a major limitation, as opposed to the radiologist's practice. To tackle this problem, a Bayesian network framework for multi-view mammographic analysis, with main focus on breast cancer detection at a patient level. The causal-independence models and context modeling over the whole breast represented as links between the regions detected by a single-view CAD system in the two breast projections. The proposed approach is implemented and tested with screening mammograms for 1063 cases of whom 385 had breast cancer. The single-view CAD system is used as a benchmark method for comparison. The results show that our multi-view modeling leads to significantly better performance in discriminating between normal and cancerous patients. It also demonstrate the potential of our multi-view system for selecting the most suspicious cases.

2.2.2 MERIT

A rule-based system is very beneficial because it contains specific and unambiguous information. The rules can be altered and updated easily whenever the need arises.

2.2.3 DEMERIT

On statistical properties, the use of rules-based has the strain and adversity that there is a possibility of deriving the similar statistical properties for some patterns of different classes, due to which there may occur a problem of incorrect recognition.

2.3 X-RAY SCATTERING IMAGE CLASSIFICATION USING DEEP LEARNING

2.3.1 DESCRIPTION

A model is proposed for feature extraction and classification of breast cancer using deep-learning and machine-learning models. Inception-v3 and Xception were used for feature extraction because of their benefits. Classification was performed using softmax, SVM, RF, KNN and ensemble. The Inception-v3 model with softmax was combined with other models such as Inception-v3-SVM, Inception-v3-RF, Inception-v3- KNN, and ensemble for the purpose of using machine learning for clinical applications related to breast cancer. Similarly, the Xception model and its combination with other models such as Xception-SVM, Xception-RF, Xception-KNN, and the Ensemble method were explored. The performance of the proposed models was addressed and compared with the existing models in breast cancer classification. The models based on the ensemble method produce highest testing accuracy and outperformed the existing models. This work could be important for the clinical applications of breast cancer analysis..

2.3.2 MERIT

The proposed system delivered 90.67% accuracy with augmented data and 87.38% without augmented data which is high compared to the previous methods which existed at that time.

2.3.3 DEMERIT

Though the sample provided atmost accuracy, it tend to fail in certain cases as Ensemble cannot help unknown differences between sample and population.

2.4 BREAST CANCER DIAGNOSIS BASED ON ENHANCED PARETO OPTIMAL AND MULTILAYER PERCEPTRON NEURAL NETWORK

2.4.1 DESCRIPTION

This proposed technique is used for breast cancer in which the classification is done using a Multilayer perceptron neural network. A non-dominated sorting genetic algorithm is used for the network assembly but local minimum is a problem of MLP. This model extract the features from the mammographic images with two views MLO and CC, ANNs are used and then focused the features of resultant mammographic images from ANNs for classification of two types of cells that are benign and malignant.

2.4.2 MERIT

This proposed model work can work for different views just as in multi view technology and as it is multi layered perceptron the accuracy is high as each layer is identified and is noted upon.

2.4.3 DEMERIT

This model cannot extract features with a well-built particular performance.

2.5 A MULTI-VIEW DEEP LEARNING ARCHITECTURE FOR CLASSIFICATION OF BREAST MICROCALCIFICATIONS [2016]

2.5.1 DESCRIPTION

The problem of differentiating between malignant and benign tumors based on

their appearance in the CC and MLO mammography views are given. Classification of clustered breast microcalcifications into benign and malignant categories is an extremely challenging task for computerized algorithms and expert radiologists. We describe a deep-learning classification method that is based on two view-level decisions, implemented by two neural networks, followed by a single-neuron layer that combines the view level decisions into a global decision that mimics the biopsy results. Our method is evaluated on a large multi-view dataset extracted from the standardized digital database for screening mammography (DDSM). Experimental results show that our network structure significantly improves on previously suggested methods.

2.5.2 MERIT

This model uses a small amount of data and achieves better accuracy. The accuracy become better by using a large amount of data.

2.5.3 DEMERIT

This model requires a high cost of implementation and as the model uses small amount of data when large data is used accuracy rate is reduced.

2.6 BATIK CLASSIFICATION USING DEEP CONVOLUTIONAL NETWORK TRANSFER LEARNING

2.6.1 DESCRIPTION

Batik fabric is one of the most profound cultural heritage in Indonesia. Hence, continuous research on understanding it is necessary to preserve it. Despite of being one of the most common research task, Batik's pattern automatic classification still requires some improvement especially in regards to invariance dilemma. Convolutional neural network (ConvNet) is one of deep learning architecture which is able to learn data representation by combining local receptive inputs, weight sharing and convolutions in order to solve invariance dilemma in image classification.

Using dataset of 2,092 Batik patches (5 classes), the experiments show that the proposed model, which used deep ConvNet VGG16 as feature extractor (transfer learning), achieves slightly better average of $89 \pm 7\%$ accuracy than SIFT and SURF-based that achieve $88 \pm 10\%$ and $88 \pm 8\%$ respectively. Despite of that, SIFT reaches around 5% better accuracy in rotated and scaled dataset.

2.6.2 MERIT

Since VGG16 extractor does not require training, it is more efficient than SIFT/SURF BoW extractor. On top of that, neural network models such as VGG16 are known to run parallelly in GPU [16] to make it even more efficient.

2.6.3 DEMERIT

Certain images in dataset often overlap each other. This condition often confuses classifier during training and causes less accurate generalization. Due to the various sources of data, the quality (resolution, noise, watermarks etc) of the data are also various. Removing low quality data and preprocessing high quality ones may produce homogeneous data and improve classifier accuracy.

2.7 A KNOWLEDGE BASED SYSTEM FOR BREAST CANCER CLASSIFICATION FUZZY LOGIC METHOD [2017]

2.7.1 DESCRIPTION

A knowledge-based system is proposed for breast cancer disease classification. The knowledge-based system uses EM, PCA, CART and fuzzy rule-based methods. WDBC and Mammographic mass datasets are used for the method evaluation. The accuracies obtained by the method are respectively 0.932 and 0.941 for WDBC and Mammographic mass datasets. Breast cancer has become a common disease around the world. Expert systems are valuable tools that have been successful for the disease diagnosis. The development of a new knowledge-based system for classification of breast cancer disease using

clustering, noise removal, and classification techniques is produced. Expectation Maximization (EM) is used as a clustering method to cluster the data in similar groups. Classification and Regression Trees (CART) is used to generate the fuzzy rules to be used for the classification of breast cancer disease in the knowledge-based system of fuzzy rule-based reasoning method. To overcome the multicollinearity issue, Principal Component Analysis (PCA) is incorporated in the proposed knowledge-based system. Experimental results on Wisconsin Diagnostic Breast Cancer and Mammographic mass datasets show that proposed methods remarkably improves the prediction accuracy of breast cancer. The knowledge-based system can be used as a clinical decision support system to assist medical practitioners in the healthcare practice.

2.7.2 MERIT

This model provides a promising accuracy in the classification task.

2.7.3 DEMERIT

This model is a hybrid intelligent system and a non-incremental which is quite expensive for implementation.

CHAPTER 3

SYSTEM SPECIFICATION

3.1 SYSTEM REQUIREMENTS

3.1.1 SOFTWARE REQUIREMENTS

Operating system	:	Windows 7
Coding Language	:	PYTHON
Tool	:	Google Colab / MATLAB

3.1.2 COLAB'S SPECIFICATION

System	:	Google Colab
Hard Disk	:	128 GB
RAM	:	8 GB

3.2 SOFTWARE DESCRIPTION

3.2.1 ABOUT GOOGLE COLAB

Colab is a free Jupyter notebook environment that runs entirely in the cloud. Most importantly, it does not require a setup and the notebooks that you create can be simultaneously edited by your team members - just the way you edit documents in Google Docs. Colab supports many popular machine learning libraries which can be easily loaded in your notebook. Typical uses include:

- Write and execute code in Python
- Import/Save notebooks from/to Google Drive

- Import/Publish notebooks from GitHub
- Import external datasets from Kaggle
- Integrate PyTorch, TensorFlow, Keras, OpenCV
- Free Cloud service with free GPU

3.2.2 ABOUT PYTHON

Python is an interpreted, high-level and general-purpose programming language. Python's design philosophy emphasizes code readability with its notable use of significant indentation. Its language constructs and object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects. Python is dynamically-typed and garbage-collected. It supports multiple programming, including structured (particularly, procedural), object oriented and functional programming. Python is often described as a "batteries included" language due to its comprehensive standard library.

3.2.3 CHARACTERISTICS OF PYTHON

- Easy to code
- High Level language
- Free and Open Source
- Integrated easily with other languages such as C, C++
- Portable language as it can be run on any platform such as Windows, Linux, Unix.
- Large standard library which provides a rich set of module and functions.
- Graphical User interfaces can be made using a module such as PyQt5, PyQt4, wx Python, or Tk in python.

CHAPTER 4

PROJECT DESIGN

4.1 OBJECTIVE

The main objective is to classify the histopathological images of breast cancer as positive or negative. A model is proposed for the classification of breast cancer on the histopathological images by using a novel deep learning method known as “Convolutional Neural Network”.

4.2 FLOW CHART

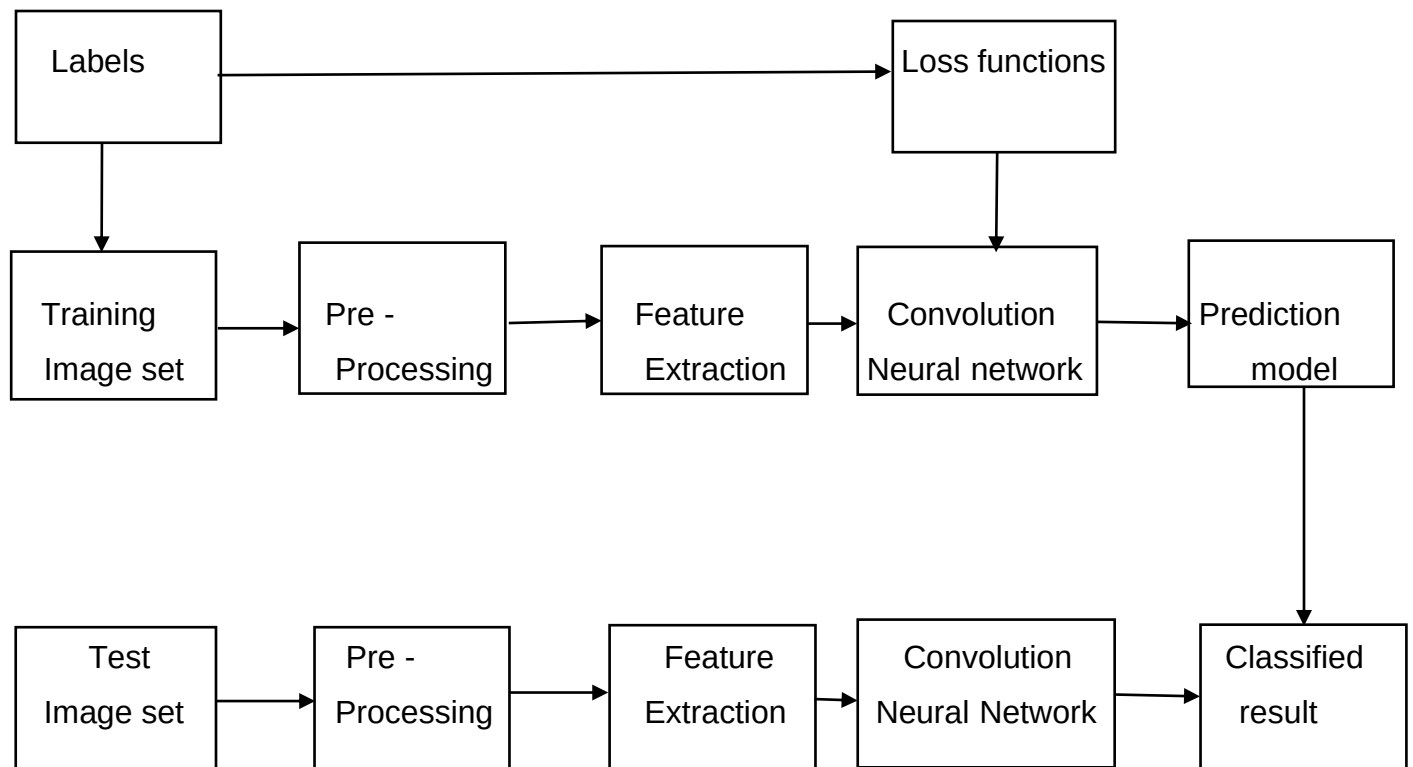


Figure 4.1 WORKFLOW OF THE PROCESS

4.3 MODULE DESCRIPTION OF THE MODEL

In deep learning, a convolutional neural network (CNN) is a type of deep neural networks, which deals with the set of data to extract information about that data. In CNN, the neural networks are trained using a heavy set of data so that the CNN can extract the feature of given input. When the input is given, first image preprocessing is done then the feature extraction occurs on the basis of set of data stored. The classification of data is done and output is shown as the result. There are three modules used: i) Pre-Processing, ii) Training the model and iii) Testing the model.

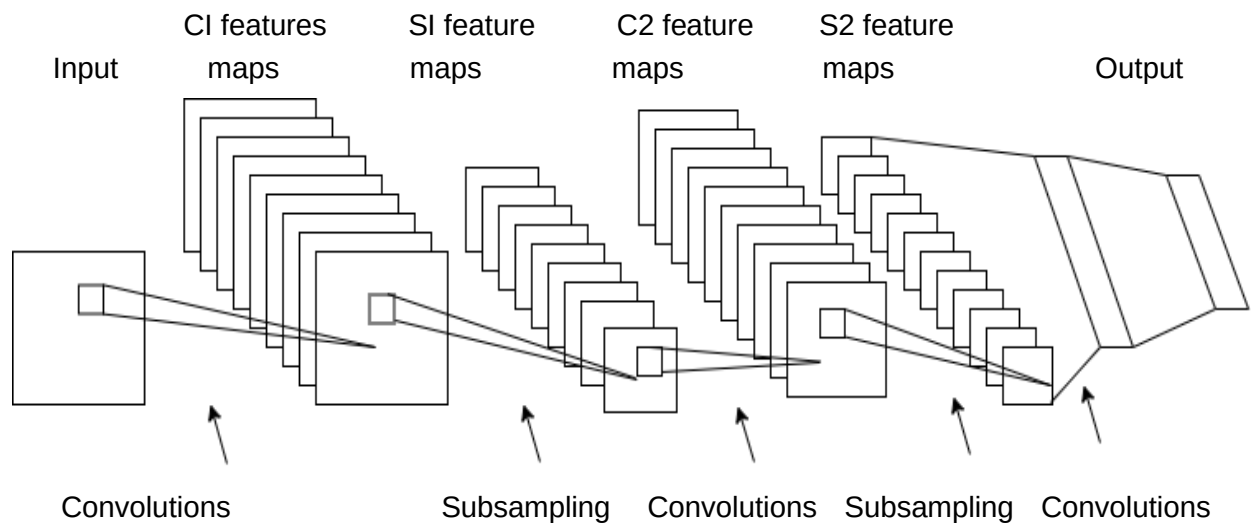


Figure 4.2 CNN ARCHITECTURE

4.3.1 PRE-PROCESSING

The pre-processing steps were conducted with resizing, patch, and augmentation. The first pre-processing step normalizes the size of the input images. Almost all the radiographs were rectangles of different heights and too large (median value of matrix size $\geq 1,800$). Accordingly, we resized all images to a standardized 32×32 pixel square, through a combination of preserving their aspect ratios and using

zero-padding. The investigation of deep learning efficiency depends on the input data. Therefore, in the second processing step is where input images are pre-processed by using a patch (a cropped part of each image). A patch was extracted using a bounding box so that it contains sufficient maxillary sinus segmentation for analysis. Finally, data augmentation was conducted for just the training dataset, using mirror images that were reversed left to right and rotated -30 , -10 , 10 , and 30 degrees.

4.3.2 TRAINING THE MODEL

After the pre-processing stage the data were sent to the training phase where the dataset is first divided into two classes namely cancerous and non-cancerous. The number of cancerous images should be equal to number of non-cancerous images to maintain a balanced dataset. The model training includes classification report that will save the model. After saving the model, it will be trained with 12 epoch (an epoch is the date and time relative to which a computer's clock and timestamp values are determined).

A CNN usually takes an order 3 tensor as its input. Higher order tensor inputs, however, can handle by CNN in similar fashion. The input then sequentially goes through a series of processing. One processing step is usually called a layer, which could be a convolutional layer, pooling layer, normalization layer, fully connected layer, loss layer etc... Here we have used CNN with the following layers:

Input layer: In this layer, an image is given as an input and produces output which is used to feed the Convolutional layers. In our example, input is of (32×32) is taken into consideration and the number of channel of image is 3 for RGB. x

Convolutional layers: In this layer, input image is convolved with a set of learnable filters, which produces a feature map corresponding to each image in the output image. In our model, there are six Convolutional layers. For first three layers size of the kernels is of 5×5 and padding is set to zero and the stride is set to two and for next three layers, we have kept the size of kernel to be 3×3 .

Fully Connected Layer or inner product layer: In this layer neurons are fully connected to each other and producing the result. Here input is simply treated as a vector and produce an output in the form of a single vector. Two inner product layers. The last layer is a fully connected layer where a softmax layer is used to classify the input image. Total number of classes used in our case is two, one for benign and other for malignant.

Pooling Layer: Convolutional layers bring out the features of images with precise positions. If the positions change, even a small amount for any reason, the feature maps will be different. To overcome this problem, the down sampling process must be done at the output of every convolutional layer. With convolutional layers, down sampling can be done by changing the convolution's phase across the image. A more acceptable and common method is to use a pooling layer. Using this process, outputs will be more accurate.

The abstract description of the CNN structure:

$$x^1 \rightarrow \boxed{w^1} \rightarrow x^2 \rightarrow \dots \rightarrow x^{L-1} \rightarrow \boxed{w^{L-1}} \rightarrow x^L \rightarrow \boxed{w^L} \rightarrow z$$

The above equation illustrates how CNN runs layer by layer in a forward pass. The input is x^1 , usually an image (order 3 tensor). It goes through the processing in the first layer, which is the first box. We denote the parameters involved in the first layer's processing collectively as tensor w^1 . The output of the first layer x^2 , which also act as the input to the second layer processing. The process proceeds till all the layers in the CNN has been finished which outputs x^L . One additional layer, however, is added for backward error propagation in CNN. Commonly used strategy is to output x^L as a C dimensional vector, whose i th entry encodes the prediction. In general, x^L may have other forms of interpretation. The last layer will be a loss layer if t is the corresponding target value for the input x^1 , then the cost or loss function can be predicted as,

$$z = \frac{1}{2} ||(t - x^L)(t - x^L)||$$

The CNN structure for the Breast cancer classification is given below:

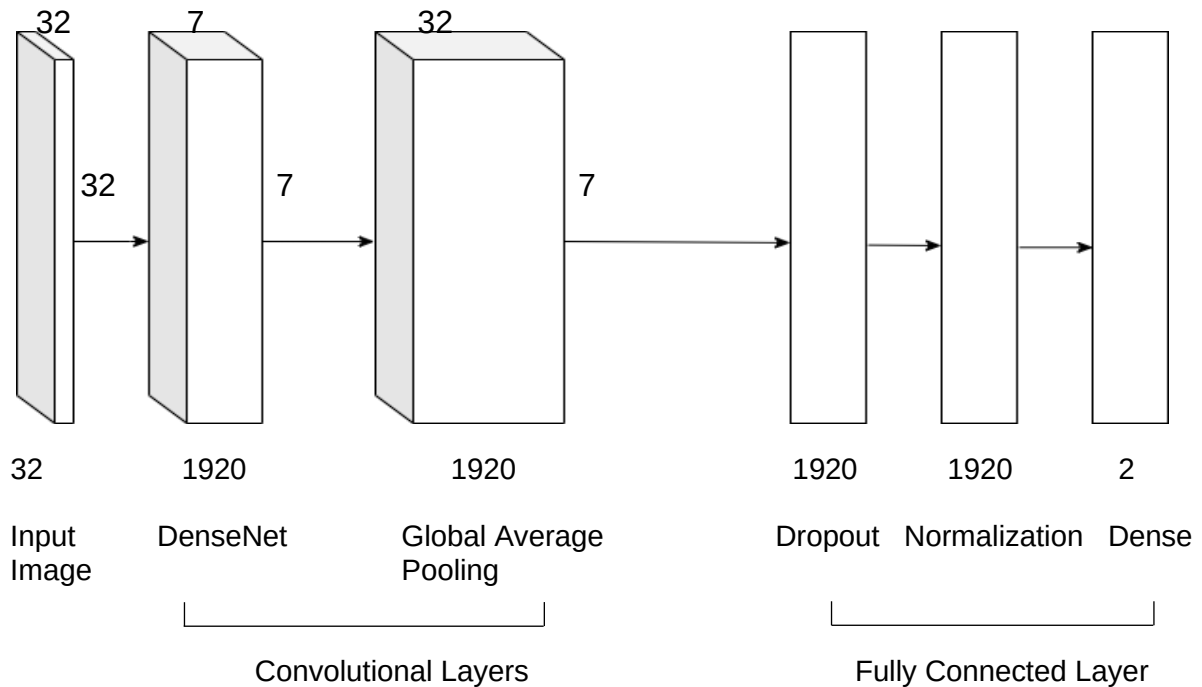


Figure 4.3 WORKING EXAMPLE OF CNN ARCHITECTURE

The input image is given as 32x32 patch image. The DenseNet layer is the n number of convolutional layers which creates convolution kernel that is convolved with the layer input to produce a tensor of outputs. Global Average Pooling layer is where, Max pooling operation for 2D spatial data occurs. Down samples the input along its spatial dimensions (height and width) by taking the maximum value over an input window (of size defined by pool size) for each channel of the input. Normalization is converting the data into a 1-dimensional array for inputting it to the next layer. The dense layer is a neural network layer that is connected deeply, which means each neuron in the dense layer receives input from all neurons of its previous layer. Normalization is converting the data into a 1-dimensional array for inputting it to the next layer. The Dropout layer randomly sets input units to 0 with a frequency of rate at each step during training time, which helps prevent overfitting.

Layer (type)	Output Shape	Param
densenet201 (Functional)	(None, 7, 7,1920)	18321984
global_average_pooling2d	(None,1920)	0
dropout (Dropout)	(None, 1920)	0
batch_normalization	(None, 1920)	0
dense (Dense)	(None, 2)	3842

Figure 4.4 MODEL SUMMARY

Total params: 18,333,506

Trainable params: 18,100,610

Non-trainable params: 232,896

Accuracy is one metric for evaluating classification Models. The model output will give the accuracy in the range of 92-95% which is calculated as by using the formula given below.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}$$

where, TP is True Positive; TN is True Negative; FP is False Positive;and FN is False Negative.

4.3.3 TESTING THE MODEL

For the testing stage, same preprocessing steps are applied on the testing data which were used in training. Image patches are created in the same fashion as in the training and these patches are then passed to the trained model. The patches from the histopathology images used in the testing stage are completely unseen

for the model. The model classified the patches as either a positive sample (cancer) or as a negative sample (background).

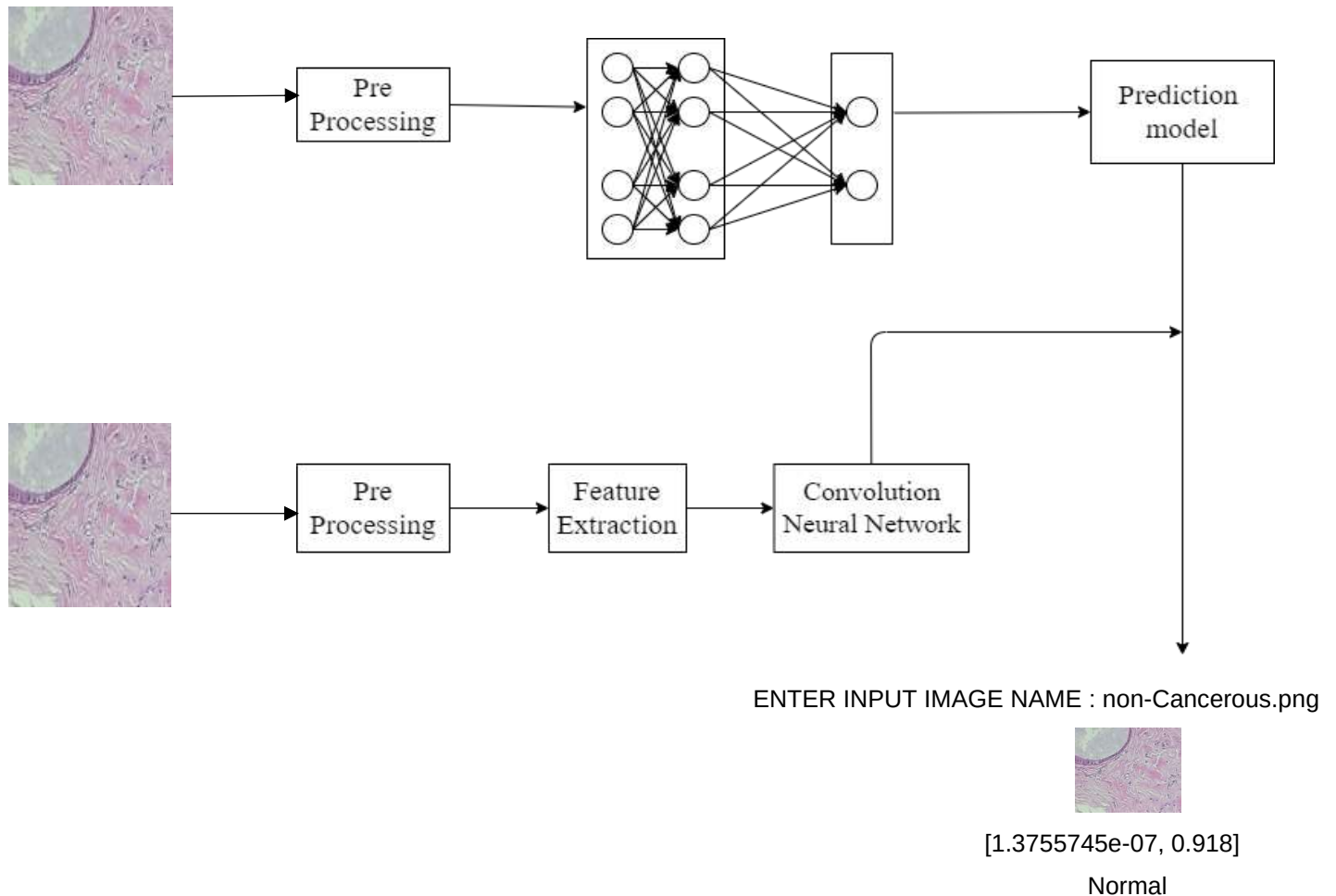


Figure 4.5 REPRESENTATION OF TRAINING AND TESTING THE MODEL

4.4 DATA AUGMENTATION TECHNIQUE

Data augmentation is an effective and widely used tool to avoid the overfitting problem by creating additional data. More complex systems using deep neural network have low bias but generate high variance. It means that these systems overfit the training data and will demonstrate bad performance on test data or on data that had not seen before. It would result in greater errors in prediction.

Therefore, the increased diversity from data augmentation decreases the model's variance by improving it at generalizing. The proposed system uses Gaussian Mixture Models (GMMs) for modeling and classification. Gaussian Mixture Models (GMMs) assume that there are a certain number of Gaussian distributions, and each of these distributions represent a cluster. Hence, a Gaussian Mixture Model tends to group the data points belonging to a single distribution together. Gaussian Mixture Models use the soft clustering technique for assigning data points to Gaussian distributions.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

Implementation is the most crucial stage in achieving a successful system and giving the users confidence that new system is effective and workable. Implementation of this project refers to the installation of the packages in its real environment to the full satisfaction of the users and operations of the system. Testing is done individually at the time of development using the data and verification is done the way specified in the program specification. In short, implementation constitutes all activities that are required to put an already tested and completed package into operation. The success of any information system lies in its successful implementation. System implementation is the stage in the project where the theoretical design is turned into a working system. The most critical stage is achieving a successful system and in giving confidence on the new system for the user that it will work efficiently and effectively. The existing system was long time process.

5.1.1 IMPORT LIBRARIES

```
import json
import math
import os
import cv2
from PIL import Image
import numpy as np
from keras import layers
from tensorflow.keras.applications import ResNet50, MobileNet, DenseNet201, InceptionV3, NASNetLarge, InceptionResNetV2, NASNetMobile
```



```

from keras.callbacks import Callback, ModelCheckpoint, ReduceLROnPlateau, TensorBoard
from keras.preprocessing.image import ImageDataGenerator
from keras.utils.np_utils import to_categorical
from keras.models import Sequential
from tensorflow.keras.optimizers import Adam
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import cohen_kappa_score, accuracy_score
import scipy
from tqdm import tqdm
import tensorflow as tf
from keras import backend as K
import gc
from functools import partial
from sklearn import metrics
from collections import Counter
import json
import itertools
%matplotlib inline

```

5.1.2 DATA PROCESSING

```

def Dataset_loader(DIR, RESIZE, sigmaX=10):
    IMG = []
    read = lambda imname: np.asarray(Image.open(imname).convert("RGB"))
    for IMAGE_NAME in tqdm(os.listdir(DIR)):
        PATH = os.path.join(DIR, IMAGE_NAME)

```

```

_, ftype = os.path.splitext(PATH)
if ftype == ".png":
    img = read(PATH)
    img = cv2.resize(img, (RESIZE,RESIZE))
    IMG.append(np.array(img))
return IMG

benign_train = np.array(Dataset_loader('data/train/benign',224))
malign_train = np.array(Dataset_loader('data/train/malignant',224))
benign_test = np.array(Dataset_loader('data/validation/benign',224))
malign_test = np.array(Dataset_loader('data/validation/malignant',224))

```

5.1.3 TRAINING AND EVALUATING THE MODEL

```

X_train = np.concatenate((benign_train, malign_train), axis = 0)
Y_train = np.concatenate((benign_train_label, malign_train_label), axis = 0)
X_test = np.concatenate((benign_test, malign_test), axis = 0)
Y_test = np.concatenate((benign_test_label, malign_test_label), axis = 0)
s = np.arange(X_train.shape[0])
np.random.shuffle(s)
X_train = X_train[s]
Y_train = Y_train[s]
s = np.arange(X_test.shape[0])
np.random.shuffle(s)
X_test = X_test[s]
Y_test = Y_test[s]
Y_train = to_categorical(Y_train, num_classes= 2)
Y_test = to_categorical(Y_test, num_classes= 2)
x_train, x_val, y_train, y_val = train_test_split(
    X_train, Y_train,

```

```

        test_size=0.2,
        random_state=11
    )

```

5.1.4 DATA GENERATOR

```

w=60
h=40
fig=plt.figure(figsize=(15, 15))
columns = 4
rows = 3
for i in range(1, columns*rows +1):
    ax = fig.add_subplot(rows, columns, i)
    if np.argmax(Y_train[i]) == 0:
        ax.title.set_text('Benign')
    else:
        ax.title.set_text('Malignant')
    plt.imshow(x_train[i], interpolation='nearest')
plt.show()
BATCH_SIZE = 1
train_generator = ImageDataGenerator(
    zoom_range=2,
    rotation_range = 90,
    horizontal_flip=True,
    vertical_flip=True,
)

```

5.1.5 MODEL DEFINITION

```

def build_model(backbone, lr=1e-4):
    model = Sequential()

```

```

model.add(backbone)
model.add(layers.GlobalAveragePooling2D())
model.add(layers.Dropout(0.5))
model.add(layers.BatchNormalization())
model.add(layers.Dense(2, activation='softmax'))
model.compile(
    loss='binary_crossentropy',
    optimizer=Adam(lr=lr),
    metrics=['accuracy']
)

return model
K.clear_session()
gc.collect()
resnet = DenseNet201(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)

model = build_model(resnet ,lr = 1e-4)
model.summary()

```

5.1.6 TRAINING DATASET

```

history = model.fit_generator(
    train_generator.flow(x_train, y_train, batch_size=BATCH_SIZE),
    steps_per_epoch=x_train.shape[0] / BATCH_SIZE,
    epochs=20,
    validation_data=(x_val, y_val),
    callbacks=[learn_control, checkpoint]
)

```

5.1.7 PLOTTING THE GRAPH

```
with open('history.json', 'w') as f:
    json.dump(str(history.history), f)
    history_df = pd.DataFrame(history.history)
    history_df[['acc', 'val_acc']].plot()
    history_df = pd.DataFrame(history.history)
    history_df[['loss', 'val_loss']].plot()
```

5.1.8 IMAGE PREDICTION

```
CATEGORIES = ["Normal","Cancer"]
image =input("ENTER INPUT IMAGE NAME: ")
from google.colab.patches import cv2_imshow
x=cv2.imread(image,1)
cv2_imshow(x)
IMG = []
read = lambda imname: np.asarray(Image.open(imname).convert("RGB"))
img = read(image)
img = cv2.resize(img, (224,224))
IMG.append(np.array(img))
prediction = model.predict(np.array(IMG))
prediction = list(prediction[0])
print(prediction)
print(CATEGORIES[prediction.index(max(prediction))])
```

5.2 SAMPLE OUTPUT

5.2.1 ACCURACY AND LOSS GRAPH

The accuracy and loss graph is drawn against Number of epoch in the given model. The Accuracy graph drawn with Number of Epoch in x-axis, and Percentage of Accuracy in y-axis. Similarly the Loss Graph is drawn with Number of Epoch in y-axis and Percentage of Loss in y-axis. The graphs are given below:

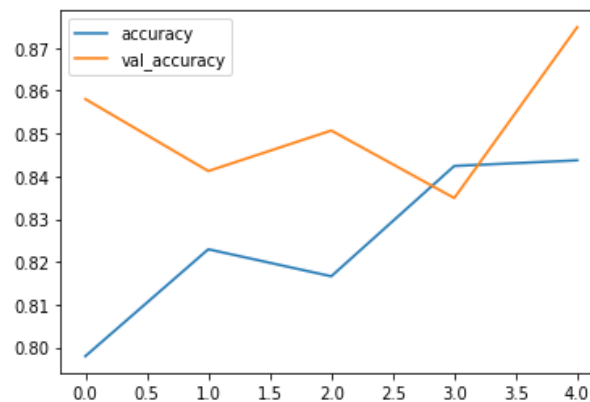


Figure 5.1 ACCURACY GRAPH

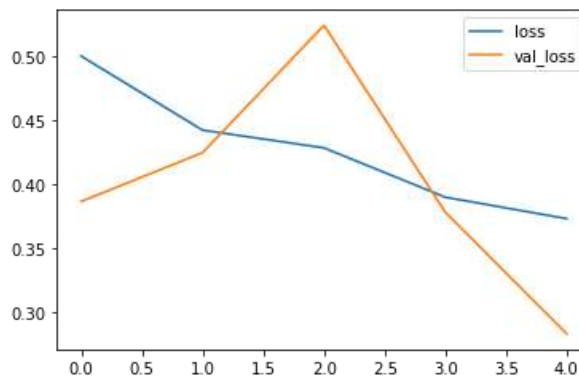


Figure 5.2 LOSS GRAPH

5.2.2 PREDICTION

An histopathological image is given as input image. The output is displayed with the input image as cancer or normal based on cancerous or non-cancerous respectively.

5.2.2.1 CANCEROUS PATCH

An input image, 2.png is given as input and the output is displayed with the input image as **Cancer** as it is a **Cancerous** patch.



FIGURE 5.3 INPUT IMAGE

ENTER INPUT IMAGE NAME: 2.png



[1.3744735e-06, 0.957]

Cancer

FIGURE 5.4 OUTPUT

5.2.2.1 NON - CANCEROUS PATCH

An input image, 3.png is given as input and the output is displayed with the input image as **Normal** as it is a **Non-Cancerous** patch.



FIGURE 5.5 INPUT IMAGE

ENTER INPUT IMAGE NAME: 3.png



[1.3755745e-07, 0.918]

Normal

FIGURE 5.6 OUTPUT

CHAPTER 6

CONCLUSION

The goal of our work was to detect and classify the cancerous and non-cancerous histopathological images using Deep learning. A novel method of deep learning technique known as Convolutional Neural Network was proposed. The proposed method includes three phases namely, Pre-processing phase, Testing phase and training phase. The histopathological images used in the process is in the form of RGB. The Region of interest (ROI) was cropped from the images and is sent to convolutional layers in the Pre-processing phase. In the Training phase, the images are trained with equal number of both cancerous and non-cancerous patches of histopathological images. The training is done with a duration of 5 epoch which is more efficient than the already existing systems. The testing is done after the training phase by giving an input image and the model will classify it as either cancerous or non-cancerous along with the display of the image. This method produced an average of 92% testing accuracy on testing samples and achieved the highest performance in the classification of Breast Cancer.

In future, we will explore and apply our work on pre-trained models trained with a larger number of layers and may also explore patch-based models with data augmentation techniques to classify Breast Cancer using histopathological images.

CHAPTER 7

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